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United States Patent	12399445
Kind Code	B2
Date of Patent	August 26, 2025
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Image forming apparatus and paper dust cleaning device

Abstract

An image forming apparatus according to the present disclosure includes a paper dust cleaning device configured to remove paper dust adhered to a surface of a conveying roller that conveys a paper sheet in a predetermined conveying direction. The paper dust cleaning device includes a paper dust removing member configured to extend in an axial direction of the conveying roller and come in contact with a surface of the conveying roller; a holder formed in a long shape in the axial direction and configured to hold the paper dust removing member; and a holder support portion configured to hold the holder so that the paper dust removing member comes in contact with the surface of the conveying roller, and configured to support the holder so as to be movable in the axial direction.

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Appl. No.:	18/600057
Filed:	March 08, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240302766 A1	Sep. 12, 2024

Foreign Application Priority Data

JP	2023-036431	Mar. 09, 2023
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Publication Classification

Int. Cl.: G03G15/02 (20060101); G03G15/16 (20060101)

U.S. Cl.:

CPC G03G15/0225 (20130101); G03G15/161 (20130101); G03G2215/00708 (20130101)

Field of Classification Search

CPC: G03G (15/0225); G03G (15/161); G03G (2215/00708); G03G (2221/0042); G03G (2221/0063); G03G (21/00)

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Background/Summary

INCORPORATION BY REFERENCE

(1) This application is based upon and claims the benefit of priority from the corresponding Japanese Patent Application No. 2023-036431 filed on Mar. 9, 2023, the entire contents of which are incorporated herein by reference.

(2) The present invention relates to an image forming apparatus that forms an image on paper, and a paper dust cleaning device.

BACKGROUND

(3) Image forming apparatuses such as printers, copiers, facsimile machines, and multifunction peripherals equipped with these functions have a printing function of forming an image on a sheet of printing paper conveyed from a paper tray by a conveying roller. When the sheet of printing paper reaches an image forming position (image transfer position), an image is formed on the printing paper. In addition, before the printing paper reaches the image forming position, the image forming apparatus may control the operation of the conveying roller to perform correction operations such as skew correction and misalignment correction of the printing paper.

(4) Moreover, conventionally, image forming apparatuses are known that include paper dust removing equipment that removes paper dust adhering to a conveying roller.

SUMMARY

(5) An image forming apparatus according to one aspect of the present invention is an image forming apparatus including a paper dust cleaning device configured to remove paper dust adhering

to a surface of a conveying roller that conveys a paper sheet in a predetermined conveying direction. The paper dust cleaning device includes a paper dust removing member configured to extend in an axial direction of the conveying roller and come in contact with a surface of the conveying roller; a holder formed in a long shape in the axial direction and configured to hold the paper dust removing member; and a holder support portion configured to hold the holder so that the paper dust removing member comes in contact with the surface of the conveying roller, and configured to support the holder so as to be movable in the axial direction.

(6) A paper dust cleaning device according to another aspect of the present invention is a paper dust cleaning device configured to remove paper dust adhering to a surface of a conveying roller that conveys a paper sheet in a predetermined conveying direction. The paper dust cleaning device includes a paper dust removing member configured to extend in an axial direction of the conveying roller and come in contact with a surface of the conveying roller; a holder formed in a long shape in the axial direction and configured to hold the paper dust removing member; and a holder support portion configured to hold the holder so that the paper dust removing member comes in contact with the surface of the conveying roller, and configured to support the holder so as to be movable in the axial direction.

(7) This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description with reference where appropriate to the accompanying drawings. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not limited to implementations that solve any or all disadvantages noted in any part of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view showing a configuration of an image forming apparatus according to an embodiment of the present invention.

(2) FIG. 2 is a schematic diagram showing an internal configuration of the image forming apparatus.

(3) FIG. 3 is a schematic diagram showing a configuration around a sheet conveying path of the image forming apparatus.

(4) FIG. 4 is a block diagram showing the configuration of the image forming apparatus.

(5) FIG. 5 is a perspective view showing a configuration around a sheet correction unit included in the image forming apparatus.

(6) FIG. 6 is a schematic diagram showing a configuration of the sheet correction unit included in the image forming apparatus.

(7) FIG. 7 is a cross-sectional view taken along cutting plane VII-VII in FIG. 6, and shows a cross-sectional structure of the sheet correction unit and a paper dust cleaning member included in the image forming apparatus.

(8) FIG. 8 is a perspective view showing a configuration of the paper dust cleaning member.

(9) FIG. 9 is a perspective view showing the configuration of the paper dust cleaning member.

(10) FIG. 10 is a cross-sectional view showing an internal structure of the paper dust cleaning member.

(11) FIG. 11 is an enlarged cross-sectional view of a slide member of the paper dust cleaning member.

(12) FIG. 12 is a diagram showing the sheet correction unit and the paper dust cleaning member, and shows a state before the paper dust cleaning member is inserted into the sheet correction unit.

(13) FIG. 13 is a diagram illustrating the sheet correction unit and the paper dust cleaning member,

and shows a state in which the paper dust cleaning member is inserted to a back portion of the sheet correction unit.

(14) FIG. 14 is a diagram showing the sheet correction unit and the paper dust cleaning member, and shows a state in which the paper dust cleaning member is pressed to a specified mounting position after the paper dust cleaning member has been inserted.

(15) FIG. 15 is a flowchart illustrating an example of a procedure of a sheet correction process executed by a control portion included in the image forming apparatus.

DETAILED DESCRIPTION

(16) Embodiments of the present invention will be described below with reference to the drawings as appropriate. The embodiments described below are merely examples of embodying the present invention, and do not limit the technical scope of the present invention.

(17) FIG. 1 is a perspective view showing a configuration of an image forming apparatus 10 according to an embodiment of the present invention. FIG. 2 is a schematic diagram showing an internal configuration of the image forming apparatus 10. In the following description, an up-down direction D1, a front-rear direction D2, and a left-right direction D3 are defined based on a state in which the image forming apparatus 10 is installed in a usable manner (the state in FIG. 1).

(18) [Image Forming Apparatus 10]

(19) As shown in FIG. 1, the image forming apparatus 10 is a multifunction peripheral capable of printing images on sheets such as printing paper, and includes various functions such as a print function, a copy function, a facsimile function, and a scan function. The image forming apparatus 10 is not limited to the above-mentioned multifunction peripheral, and may be any apparatus as long as the apparatus has a print function of printing an image on a conveyed sheet, and may be, for example, a printer, a copying machine, a FAX machine, or the like.

(20) The image forming apparatus 10 includes an image reading portion 12 and an image forming portion 14. The image reading portion 12 performs a reading process of reading an image of a document, and is provided at an upper portion of the image forming apparatus 10. The image forming portion 14 performs a process of forming a color image based on an electrophotographic method, and is provided at a lower portion of the image forming apparatus 10. In addition, a sheet discharge portion 15 is provided on a right side of the image forming portion 14.

(21) A discharge space 21 is provided above the image forming portion 14. The sheet discharge portion 15 vertically connects the image forming portion 14 and the image reading portion 12 while forming the discharge space 21 between the image forming portion 14 and the image reading portion 12.

(22) The sheet discharge portion 15 discharges the sheet after image formation to the discharge space 21. A sheet discharge port 15A (see FIG. 2) is formed on a left side surface of the sheet discharge portion 15 on the discharge space 21 side. The sheet is discharged from the sheet discharge port 15A.

(23) The image forming portion 14 includes a housing 11 as a main body of the apparatus. Inside the housing 11, various elements of the image forming portion 14 are provided. The housing 11 includes an exterior frame that covers the entire image forming portion 14 and an internal frame that supports each part of the image forming portion 14.

(24) A front cover 111 (an example of a locking member or a cover member of the present invention) is provided on a front surface of the housing 11 to open the inside of the housing 11. The front cover 111 is rotatably supported to be able to rotate toward the front by a support shaft (not shown) provided at a left end portion of the front surface of the housing 11 and extending in the up-down direction D1.

(25) FIG. 2 is a schematic diagram showing an internal configuration of the image forming apparatus 10. In FIG. 2, illustration of the image reading portion 12 is omitted.

(26) The image forming portion 14 forms a color image on a sheet such as printing paper based on a so-called tandem method. As shown in FIG. 2, the image forming portion 14 includes a plurality

of image forming units **4**, an intermediate transfer unit **5**, a laser scanning unit **13**, a secondary transfer roller **20**, a fixing device **16**, a sheet tray **18**, a sheet storing portion **27**, a feeding unit **28**, an operation display portion **17** (see FIG. **1**), a sheet conveying path **26** (hereinafter abbreviated as the conveying path **26**), a sheet conveying unit **23**, a sheet correction unit **60**, a toner container **3**, and a control portion **30** (see FIG. **4**) and the like.

(27) As shown in FIG. **2**, the sheet storing portion **27** is provided at a bottom portion of the image forming apparatus **10**. The sheet storing portion **27** stores a sheet onto which an image is transferred by the image forming unit **4**, and is formed, for example, in the shape of a tray with an open upper side. The sheet storing portion **27** is supported by the housing **11**.

(28) The feeding unit **28** takes out a plurality of sheets stacked in the sheet storing portion **27** one by one, and feeds the sheets toward the conveying path **26**. The feeding unit **28** includes a pickup roller **29** and a feeding roller pair **22**. The pickup roller **29** and the feeding roller pair **22** are provided above a right side portion of the sheet storing portion **27**.

(29) FIG. **3** is a schematic diagram showing a configuration around the conveying path **26**. The feeding roller pair **22** receives a rotational driving force from a conveying motor **51** (see FIG. **4**) to convey the sheet toward a downstream side in the conveying direction **D11**. As shown in FIG. **3**, the feeding roller pair **22** includes a drive roller **22A** that rotates by a rotational driving force transmitted from the conveying motor **51**, and a driven roller **22B** that is driven and rotated by being in contact with the drive roller **22A**. A drive transmission mechanism (not shown) that transmits the rotation of the drive roller **22A** to a pickup roller **29** is provided between the pickup roller **29** and the drive roller **22A**. The pickup roller **29** and the drive roller **22A** are connected via the drive transmission mechanism. When the drive roller **22A** is rotated by the conveying motor **51**, the pickup roller **29** is also rotated in the same direction as the drive roller **22A** by the drive transmission mechanism.

(30) When an instruction signal to start a sheet feeding operation is input to the image forming apparatus **10**, the pickup roller **29** and the feeding roller pair **22** are rotated by the rotational driving force of the conveying motor **51**, and the sheet is fed from the sheet storing portion **27** to the conveyance path **26**.

(31) The conveying path **26** is a guide path that guides the sheet fed by the feeding roller pair **22** to the sheet discharge port **15A**. As shown in FIG. **2**, the conveying path **26** is curved upward from the feeding roller pair **22** and then extends upward, passing the secondary transfer roller **20** and reaching the sheet discharge port **15A**.

(32) As shown in FIG. **2**, a side cover **112** is provided on a right side surface of the housing **11** to expose the conveying path **26** inside the housing **11**. The side cover **112** is rotatably supported by a support shaft (not shown) provided at a lower end portion of the opening on the right side surface of the housing **11** and extending in the front-rear direction **D2**. A conveying guide of the right guide surface of the conveying path **26** is integrally provided on an inner surface side of the side cover **112**. Therefore, when the side cover **112** is opened, the conveying path **26** is opened.

(33) As shown in FIG. **3**, the sheet conveying unit **23** and the sheet correction unit **60** are provided near the conveying path **26**.

(34) The sheet conveying unit **23** conveys the sheet toward a downstream side of a conveying direction **D11** by receiving a rotational driving force from the conveying motor **51** (see FIG. **4**). The sheet fed out to the conveying path **26** by the feeding unit **28** is conveyed toward the downstream side in the conveying direction **D11** by the sheet conveying unit **23**.

(35) The sheet correction unit **60** is arranged on the conveying path **26** farther on the downstream side in the conveying direction **D11** than the sheet conveying unit **23**. The sheet correction unit **60** corrects the orientation and position of the sheet that is conveyed along the conveying path **26** at an angle with respect to the conveying direction **D11**, and conveys the sheet toward the downstream side in the conveying direction **D11**. The configuration of the sheet correction unit **60** will be described later.

(36) As shown in FIG. 2, each image forming unit 4 is provided below the intermediate transfer unit 5. Each image forming unit 4 performs an image forming process to form a toner image on the surface of a transfer belt 5A based on image data input from the outside. The image forming units 4 are arranged side-by-side along the running direction of the transfer belt 5A (direction indicated by arrow D10). From the left side to the right side of the transfer belt 5A, an image forming unit 4Y for yellow, an image forming unit 4C for cyan, an image forming unit 4M for magenta, and an image forming unit 4K for black are arranged in a line in that order.

(37) Each of the image forming units 4 includes a photoconductor drum 41, a charging device 42, a developing device 44, a primary transfer roller 45, and the like. The image forming unit 4Y forms a toner image on the surface of the photoconductor drum 41 using yellow toner. The image forming unit 4C forms a toner image on the surface of the photoconductor drum 41 using cyan toner, the image forming unit 4M forms a toner image on the surface of the photoconductor drum 41 using magenta toner, and the image forming unit 4K forms a toner image on the surface of the photoconductor drum 41 using black toner. A developing process of developing the toner image on the photoconductor drum 41 is performed by a developing device 44.

(38) The intermediate transfer unit 5 includes a transfer belt 5A, a drive roller 5B, and a driven roller 5C. The transfer belt 5A is an annular belt member to which the toner images of each color formed on the photoconductor drums 41 of the image forming units 4 are transferred. The transfer belt 5A is provided above the photoconductor drums 41. The transfer belt 5A is rotatably supported by the drive roller 5B and the driven roller 5C that are provided apart from each other in the left-right direction D3. The transfer belt 5A is supported so as to span over the drive roller 5B and the driven roller 5C. When the surface of the transfer belt 5A passes between the photoconductor drums 41 and the primary transfer rollers 45, toner images are sequentially transferred to the transfer belt 5A from each photoconductor drum 41 in an overlapping manner.

(39) The laser scanning unit 13 irradiates the photoconductor drum 41 of each image forming unit 4 with a laser beam based on input image data of each color. Thus, an electrostatic latent image is formed on each photoconductor drum 41.

(40) The secondary transfer roller 20 is provided to face the drive roller 5B with the conveying path 26 extending vertically in between. The secondary transfer roller 20 performs a transfer process of transferring the toner image on the transfer belt 5A to the surface (transfer surface) of the sheet using a transfer potential applied to the secondary transfer roller 20. The sheet onto which the toner image has been transferred is conveyed to a fixing device 16.

(41) The fixing device 16 heats the toner image transferred to the sheet and fixes the toner image to the sheet, and includes a heating roller 16A and a pressure roller 16B. The sheet conveyed to the fixing device 16 is conveyed while being sandwiched between the heating roller 16A and the pressure roller 16B. During conveying, heat is transferred from the heating roller 16A to the toner image transferred to the sheet, thereby heating the toner image. Thus, the toner image is fixed on the surface of the sheet. After that, the sheet is discharged onto the sheet tray 18 by the sheet discharge portion 15.

(42) [Sheet Conveying Unit 23]

(43) As shown in FIG. 3, the sheet conveying unit 23 includes a drive roller 23A that is rotationally driven by receiving a driving force from the conveying motor 51 (see FIG. 4), and a driven roller 23B arranged in a state so as to be in contact with the outer peripheral surface of the drive roller 23A. A conveying roller pair is achieved by the drive roller 23A and the driven roller 23B.

(44) The drive roller 23A is fixed to a rotating shaft 47 extending in the front-rear direction D2 (hereinafter may be referred to as the width direction D2), and the rotating shaft 47 is rotatably supported by the internal frame of the housing 11. A driving force from the conveying motor 51 (see FIG. 4) is transmitted to the rotating shaft 47. The driving force of the conveying motor 51 is transmitted to the rotating shaft 47 via a transmission mechanism (not shown) such as a gear or a belt.

(45) The driven roller **23B** is biased toward the drive roller **23A** side by a spring **23C** with a predetermined elastic force (spring force). A bearing portion (not shown) that pivotally supports the rotating shaft **49** of the driven roller **23B** is biased by the spring **23C**. Thus, the driven roller **23B** is pressed against the drive roller **23A**. When the drive roller **23A** is rotationally driven in this state, the driven roller **23B** is driven.

(46) As shown in FIG. **3**, the conveying path **26** is provided with a leading-edge detection sensor **77**. The leading-edge detection sensor **77** detects the leading edge of the sheet conveyed over the conveying path **26**. In the conveying path **26**, the leading-edge detection sensor **77** is provided farther on the downstream side in the conveying direction **D11** than the sheet conveying unit **23**, and farther on the upstream side in the conveying direction **D11** than the sheet correction unit **60**.

(47) The leading-edge detection sensor **77** is, for example, a line sensor extending in the width direction **D2**. The line sensor includes a plurality of image sensors arranged in a row in the width direction **D2**. The leading-edge detection sensor **77** is connected to the control portion **30**. The leading-edge detection sensor **77** outputs a detection signal including image data (density data) of the leading edge of the sheet to the control portion **30**. After acquiring the detection signal from the leading-edge detection sensor **77**, the control portion **30**, based on the detection signal, determines whether the sheet has reached the detection position of the leading-edge detection sensor **77**. Since this determination method is conventionally known, a detailed description thereof will be omitted.

(48) Note that the leading-edge detection sensor **77** may be a reflective optical sensor provided at the center of the conveying path **26**. The reflective optical sensor includes a light emitting element and a light receiving element, receives reflected light emitted from the light emitting element, and outputs a detection signal according to the amount of the received light. The control portion **30** may determine whether the conveyed sheet has reached the detection position of the leading-edge detection sensor **77** based on a change point of the detection signal sent from the reflective optical sensor.

(49) As shown in FIG. **3**, in the conveyance path **26**, an edge detection sensor **78** is provided farther on the downstream side in the conveying direction **D11** than the sheet correction unit **60**.

(50) The edge detection sensor **78** is arranged farther on the downstream side in the conveying direction **D11** than the sheet correction unit **60** and farther on the upstream side in the conveying direction **D11** than the image transfer position by the secondary transfer roller **20**. The edge detection sensor **78** detects positions at both ends in the width direction **D2** of the sheet that has passed through the sheet correction unit **60**. The edge detection sensors **78** is a pair of line sensors provided at positions equally spaced from the center of the conveying path **26** to the outside in the width direction **D2**. Each line sensor is composed of a plurality of image sensors arranged in a row in the width direction **D2**. In the present embodiment, the edge detection sensor **78** is arranged so that the end portions of the sheet in the width direction **D2** pass by each line sensor.

(51) As shown in FIG. **4**, the edge detection sensor **78** is connected to the control portion **30**. The edge detection sensor **78** outputs a detection signal (density signal) including image data (density data) of both end portions of the sheet to the control portion **30**. After acquiring the detection signal from the edge detection sensor **78**, the control portion **30** determines the position of the sheet in the width direction **D2** based on the detection signal. More specifically, the control portion **30** determines whether the sheet is located at the center in the width direction **D2**, whether the sheet has shifted in any direction in the width direction **D2**, and in a case where the sheet has shifted in the width direction **D2**, determines the amount shifted (lateral shift amount). Note that, since such a determination method is conventionally known, a detailed explanation thereof will be omitted.

(52) [Sheet Correction Unit **60**]

(53) As shown in FIG. **3**, the sheet correction unit **60** is provided on the conveying path **26** farther on the downstream side in the conveying direction **D11** than the sheet conveying unit **23**. More specifically, the sheet correction unit **60** is provided between the leading-edge detection sensor **77** and the edge detection sensor **78** in the conveying path **26**.

(54) FIG. 5 is a perspective view showing the configuration around the sheet correction unit **60**, and is a view of the sheet correction unit **60** as viewed from the diagonally rear right in FIG. 3. FIG. 6 is a schematic diagram showing the configuration of the sheet correction unit **60**, and is a view of the sheet correction unit **60** as viewed from the right side in FIG. 3. Note that FIG. 5 shows a state in which the side cover **112** is removed and the registration roller **61A** and the like is exposed.

(55) In a case where the sheet conveyed by the sheet conveying unit **23** is shifted in the width direction **D2**, the sheet correction unit **60** returns the sheet to a predetermined center position and corrects the shift (lateral shift). More specifically, the sheet correction unit **60**, during the process in which the sheet that has entered the sheet correction unit **60** is conveyed while being sandwiched between the registration roller **61A** and the driven roller **61B**, which will be described later, moves the sheet in a direction (direction opposite to the shift direction) in which the lateral shift is corrected by the amount of shift in the width direction **D2** of the sheet. Thus, the lateral shift of the sheet in the width direction **D2** is corrected.

(56) As shown in FIG. 5, the sheet correction unit **60** is attached to a bracket **110** (an example of a frame support portion of the present invention) that is fixed to the housing **11**. The sheet correction unit **60** includes a movable frame **65** (an example of a roller support frame of the invention), a registration roller **61A** (an example of a conveying roller of the invention), a driven roller **61B**, and a paper dust cleaning member **80** (example of a paper dust cleaning device of the present invention, see FIG. 6). Note that the paper dust cleaning member **80** is housed inside the movable frame **65**, and thus the paper dust cleaning member **80** is not shown in FIG. 5.

(57) The bracket **110** is fixed to the housing **11**. The movable frame **65** is attached to the bracket **110**. In the present embodiment, the movable frame **65** is formed in a long shape in the width direction **D2**. In addition, the movable frame **65** is supported by the bracket **110** of the housing **11** so as to be movable in the width direction **D2**. Note that as a support mechanism for supporting the movable frame **65** by the bracket **110** so as to be movable in the width direction **D2**, it is possible to apply, for example, a well-known slide rail mechanism composed of a rail and a guide, or other well-known support mechanisms.

(58) As shown in FIG. 5, a rack **71** is formed at a lower end portion of the movable frame **65**.

(59) A driven pulley **73** having a pinion gear **72** is rotatably provided on the bracket **110**. A pinion gear **72** is engaged with the rack **71**. In addition, a drive pulley **75** is rotatably provided on the bracket **110**. The drive pulley **75** is connected to the driven pulley **73** by a belt **74**.

(60) The bracket **110** is provided with a correction motor **52** (an example of a drive portion of the present invention, see FIG. 4) for supplying a rotational driving force to the drive pulley **75**. A drive pulley **75** is attached to an output shaft of the correction motor **52**. Therefore, when the correction motor **52** is rotationally driven, the drive pulley **75** receives a rotational driving force and rotates, and the rotational driving force is transmitted to the driven pulley **73** via the belt **74**.

(61) In the present embodiment, when the correction motor **52** is driven and controlled by the control portion **30** to rotate forward or rearward, the movable frame **65** slides and moves in either direction (forward direction or rearward direction) in the width direction **D2**.

(62) As shown in FIG. 6, a protruding piece **66** that protrudes rearward is provided at a rear end portion of the movable frame **65**. In addition, an optical sensor **76** capable of detecting the protruding piece **66** is provided on a rear side of the movable frame **65**. The optical sensor **76** is attached to the housing **11** or the bracket **110**, for example. The position of the protruding piece **66** is detected by the optical sensor **76**. As will be described later, by moving the movable frame **65** to the rear side together with the registration roller **61A**, the protruding piece **66** enters the detection area of the optical sensor **76** and is detected by the optical sensor **76**. Thus, the movable frame **65** is arranged at a predetermined initial position. That is, the position of the movable frame **65** when the protrusion piece **66** is detected by the optical sensor is the initial position.

(63) In the present embodiment, when the main power supply of the image forming apparatus **10** is

turned ON from OFF, the control portion **30**, based on the initial position where the protruding piece **66** is detected by the optical sensor **76**, moves the movable frame **65** by a predetermined amount to one side (front side) in the width direction **D2**, and moves the movable frame **65** and the registration roller **61A** to a predetermined home position.

(64) The registration roller **61A** is a roller member that is long in the width direction **D2**. The registration roller **61A** is rotatably supported by the movable frame **65**. The movable frame **65** is provided with a pair of support plates **67**, **68** spaced apart from each other by a predetermined distance in the width direction **D2**. The predetermined interval is a length that allows the sheet to be conveyed between the pair of support plates **67**, **68**. The registration roller **61A** is provided with rotating shafts **61A1** at both ends in the axial direction, and the rotating shafts **61A1** are rotatably supported by the support plates **67**, **68**.

(65) An input gear **63** is attached to a rotating shaft **61A1** on one side (rear side) of the registration roller **61A**. A gear transmission mechanism **64** including the input gear **63** is provided on a rear side of the sheet correction unit **60**. A conveying motor **53** such as a stepping motor is provided on the bracket **110** or the housing **11**. By inputting the rotational driving force of the conveying motor **53** to the gear transmission mechanism **64**, the registration roller **61A** is rotationally driven, and the sheet is conveyed in the conveying direction **D11**.

(66) The driven roller **61B** is rotatably supported by the conveying guide of the side cover **112**. When the side cover **112** is closed with respect to the housing **11**, the driven roller **61B** contacts the outer peripheral surface of the registration roller **61A** with a predetermined elastic force by the spring **62** (see FIG. 3).

(67) A conveying roller pair **61** is achieved by the registration roller **61A** and the driven roller **61B**. This conveying roller pair **61** conveys the sheet toward the image transfer position on the downstream side in the conveying direction **D11**.

(68) [Paper Dust Cleaning Member **80**]

(69) FIG. 7 is a diagram showing a cross-sectional structure of the sheet correction unit **60** and the paper dust cleaning member **80**. FIG. 8 to FIG. 11 are diagrams showing the structure of the paper dust cleaning member **80**. The paper dust cleaning member **80** removes paper dust adhered to a surface of the registration roller **61A** that conveys the sheet in the conveying direction **D11**, and is mounted to the movable frame **65** of the sheet correction unit **60** as shown in FIG. 7.

(70) As shown in FIG. 7, the movable frame **65** is integrally provided with a storage portion **81** (an example of a holder support portion of the present invention) to which the paper dust cleaning member **80** is mounted. The storage portion **81** is configured to support the paper dust cleaning member **80** so as to be movable in the front-rear direction **D2**. Here, the front-rear direction **D2** is a direction that coincides with the axial direction or longitudinal direction of the registration roller **61A**. In the present embodiment, in a case where the paper dust cleaning member **80** is mounted to a mounting position set in the storage portion **81**, the storage portion **81** holds the paper dust cleaning member **80** so that the paper dust removing portion **90** of the paper dust cleaning member **80** contacts the surface of the registration roller **61A**. Note that FIG. 7 shows a state in which the paper dust cleaning member **80** is mounted at the mounting position and the paper dust removing portion **90** is in contact with the surface of the registration roller **61A**.

(71) The storage portion **81** is provided below the movable frame **65**, and more specifically, is provided below the registration roller **61A**. The storage portion **81** includes an insertion opening **82** formed on a wall surface on the front side of the movable frame **65**, a support plate **83** extending rearward from a lower edge portion **82A** of an insertion opening **82**, and a back wall **84** located on a rear side of the storage portion **81** (an example of a contact portion of the present invention). A space defined by the support plate **83** and the surface of the registration roller **61A** in a portion from the insertion opening **82** to the back wall **84** is a storage chamber **81A** (storage space) in which the paper dust cleaning member **80** is stored.

(72) When the paper dust cleaning member **80** is horizontally inserted into the insertion opening

82, a lower portion of the paper dust cleaning member **80** is slidably supported by an upper surface **83A** of the support plate **83**. That is, the paper dust cleaning member **80** is supported by the support plate **83** so as to be movable in the front-rear direction **D2**. When the paper dust cleaning member **80** is pushed rearward, a tip-end portion **91A** of the paper dust cleaning member **80** comes into contact with the back wall **84**, and the paper dust cleaning member **80** is arranged at the mounting position.

(73) As shown in FIG. **8** and FIG. **9**, the paper dust cleaning member **80** includes the paper dust removing portion **90** (an example of the paper dust removing member of the present invention) and a holder **91** that holds the paper dust removing portion **90**. The paper dust cleaning device of the present invention is achieved by the above-described storage portion **81**, the paper dust removing portion **90**, and the holder **91**.

(74) Conventional paper dust removing members used for removing paper dust are formed of sponge-like or brush-like members. Therefore, in a case where a conveying roller such as the registration roller **61A** moves in the axial direction, a conventional paper dust removing member may be deformed in the direction of movement, resulting in a decrease in paper dust removing ability. In addition, when the conveying rollers such as the registration roller **61A** are frequently moved in the axial direction during skew correction or misalignment correction, the repetition of normal and deformed positions may damage or shorten the life span of the conventional paper dust removing member. Moreover, due to repeated changes in posture, there is a risk that removed paper dust existing on the surface of or inside of the conventional paper dust removing member may be scattered and adhere to conveying rollers such as the registration roller **61A** or adhere to the paper.

(75) With the image forming apparatus **10** of the present embodiment, it is possible to prevent deterioration of the paper dust removing portion **90** (paper dust removing member) due to the movement in the axial direction of the conveying rollers such as the registration roller **61A**, and prevent scattering of the removed paper dust.

(76) The holder **91** is a long bar-shaped member that is longer than the length in the axial direction of the registration roller **61A**. The holder **91** is, for example, a resin molded product made of synthetic resin. The paper dust cleaning member **80** is mounted in the storage portion **81** so that the longitudinal direction **D21** of the holder **91** and the axial direction of the registration roller **61A** coincide.

(77) The paper dust removing portion **90** is held on an upper surface of the holder **91**. The paper dust removing portion **90** is, for example, a columnar member with a rectangular cross section, with one side surface being in contact with the upper surface of the holder **91**. The paper dust removing portion **90** is made of, for example, a soft material, such as a sponge member, a brush-like member, a raised member, a nonwoven fabric member, a foam member, or the like. Specific examples of the paper dust removing portion **90** include, for example, polyurethane foam, felt, cork, and the like.

(78) A protrusion **92** is formed on the tip-end portion **91A** of the holder **91**. A through hole **84A** (see FIG. **7**) into which the protrusion **92** can be inserted is formed in the back wall **84** of the storage portion **81**. When the paper dust cleaning member **80** is mounted at the mounting position of the storage portion **81**, the protrusion **92** is inserted into the through hole **84A**. Thus, the tip-end portion **91A** of the holder **91** is positioned in a direction perpendicular to the longitudinal direction **D21**. In addition, the tip-end portion **91A** of the holder **91** is restricted by the back wall **84** so as not to move further rearward. Note that the protrusion length of the protrusion **92** is formed to be longer than the maximum correction amount in the width direction **D2** by the sheet correction unit **60** (the maximum sliding amount of the movable frame **65**).

(79) As shown in FIG. **10**, a slide member **93** is attached to a base end portion **91B** on one side (front side) in the longitudinal direction **D21** of the holder **91**. The slide member **93** is slidably attached to the base end portion **91B** in the longitudinal direction **D21** while forming a storage space **94** between the slide member **93** and an end surface of the base end portion **91B**.

(80) The slide member **93** has a cylindrical portion **95** that defines the storage space **94** therein. The base end portion **91B** is inserted into an opening on one side of the cylindrical portion **95**. A plurality of slot holes **95A** that are long in the longitudinal direction **D21** are formed in the outer peripheral surface of the cylindrical portion **95**. The base end portion **91B** is provided with a plurality of engaging hooks **96** that protrude from an end surface of the base end portion **91B**. With the base end portion **91B** inserted into the cylindrical portion **95**, hook portions **96A** at the tip ends of the engaging hooks **96** are inserted into the slot holes **95A**. The slot holes **95A** are longer in the longitudinal direction **D21** than the hook portions **96A**. Therefore, with the hook portions **96A** engaged with the slot holes **95A**, the slide member **93** can slide within a predetermined range with respect to the base end portion **91B**.

(81) A spring **97** (an example of the elastic member of the present invention) is attached to the storage space **94**. The spring **97** provides an elastic force that biases the slide member **93** and the holder **91** in directions away from each other. The spring **97** is, for example, a coil spring, and is stored in the storage space **94** in a compressed state so as to generate the elastic force.

(82) As shown in FIG. 7, the storage portion **81** is provided with a first inclined guide **101** (an example of the first guide portion of the present invention). In a case where the paper dust cleaning member **80** is pulled out to the front side from the mounted state in which the paper dust cleaning member **80** is mounted in the storage portion **81**, the first inclined guide **101** guides the paper dust cleaning member **80** according to the pulling-out operation to a predetermined separated position (the position shown in FIG. 13). The separated position is a position where the paper dust cleaning member **80** is separated from the surface of the registration roller **61A** to generate a gap between the surface of the registration roller **61A** and the paper dust removing portion **90**.

(83) In addition, the first inclined guide **101** guides the paper dust cleaning member **80** from the separated position to a predetermined contact position (the position shown in FIG. 7) in the process of mounting the paper dust cleaning member **80** in the storage portion **81**. The contact position is a position where the paper dust removing portion **90** approaches the surface of the registration roller **61A** from the separated position and comes into contact with the surface of the registration roller **61A**.

(84) The first inclined guide **101** is integrally formed with the support plate **83**. The first inclined guide **101** is provided at a rear end portion of the support plate **83**. The first inclined guide **101** is an inclined surface inclined rearward and upward from an upper surface **83A** of the support plate **83**. The inclination angle of the first inclined guide **101** may be set to any angle, for example, within the range of 0 degrees to 45 degrees.

(85) In the support plate **83**, a flat support portion **83B** is formed farther on the rear side than the first inclined guide **101**. With the paper dust cleaning member **80** mounted to the storage portion **81**, a lower end portion **91A1** of the tip-end portion **91A** is supported by the flat support portion **83B**.

(86) In addition, as shown in FIG. 7, the holder **91** is provided with a second inclined guide **102** (an example of a second guide portion of the present invention). In a case where the paper dust cleaning member **80** is pulled out to the front side from the mounted state in which the paper dust cleaning member **80** is mounted in the storage portion **81**, the second inclined guide **102** guides the paper dust cleaning member **80** according to the pulling-out operation to the above-mentioned separated position.

(87) Moreover, the second inclined guide **102** guides the paper dust cleaning member **80** from the separated position to the contact position in the process of mounting the paper dust cleaning member **80** in the storage portion **81**.

(88) The second inclined guide **102** is integrally formed with the holder **91**. The second inclined guide **102** is provided on the base end portion **91B** side of the holder **91**. The second inclined guide **102** is an inclined surface inclined rearward and upward from a lower surface **91B1** of the base end portion **91B**. The angle of inclination of the second inclined guide **102** is set to the same angle as

the angle of inclination of the first inclined guide **101**.

(89) The work of mounting the paper dust cleaning member **80** in the storage portion **81** is performed while the power supply of the image forming apparatus **10** is turned OFF. When the front cover **111** (see FIG. **1**) of the image forming apparatus **10** is opened, the insertion opening **82** is exposed (see FIG. **12**).

(90) When the paper dust cleaning member **80** is horizontally inserted into the insertion opening **82** and pressed toward the back side, the tip-end portion **91A** reaches the vicinity of the back wall **84** (see FIG. **13**). At this time, the lower end portion **91A1** of the tip-end portion **91A** of the holder **91** is arranged at a lower end portion **101A** of the first inclined guide **101**. In this state, the second inclined guide **102** of the holder **91** has not yet come into contact with the lower edge portion **82A** of the insertion opening **82**.

(91) After that, when the front cover **111** of the image forming apparatus **10** is closed, the back surface of the front cover **111** comes into contact with a front end portion **93A** (see FIG. **7**) of the slide member **93** during the closing process, and the slide member **93** is pushed to the back side (rear side) (see FIG. **7**). Thus, the paper dust cleaning member **80** approaches the back side. As the paper dust cleaning member **80** approaches, the lower end portion **91A1** of the tip-end portion **91A** of the holder **91** rides up on the slope of the first inclined guide **101**, and the tip-end portion **91A** moves up the first inclined guide **101** (see FIG. **14**). Thus, the tip-end portion **91A** is pushed up toward the registration roller **61A** side. When the lower end portion **91A1** reaches the flat support portion **83B** and the paper dust cleaning member **80** further moves toward the back side, the protrusion **92** is inserted into the through hole **84A** of the back wall **84**, and the tip-end portion **91A** is inserted into the back wall **84**.

(92) In the process of the lower end portion **91A1** of the tip-end portion **91A** moving up the first inclined guide **101**, the second inclined guide **102** of the holder **91** comes into contact with the lower edge portion **82A** of the insertion opening **82**. Thus, the base end portion **91B** of the holder **91** is pushed up toward the registration roller **61A** side, behind the tip-end portion **91A**. Before the protrusion **92** is inserted into the through hole **84A** of the back wall **84** and the tip-end portion **91A** comes into contact with the back wall **84**, a lower surface **91B1** of the base end portion **91B** passes over the lower edge portion **82A** and rides up on the upper surface **83A** of the support plate **83**. At this time, as shown in FIG. **7**, the entire paper dust removing portion **90** comes in contact with the surface of the registration roller **61A** with uniform pressure. When the back surface of the front cover **111** catches up with the paper dust cleaning member **80** and is moved further to the back side, the contact friction force between the paper dust removing portion **90** and the registration roller **61A** causes the registration roller **61A** and the movable frame **65** to move toward the initial position on the back side.

(93) During the closing process in which the front cover **111** is closed, when the paper dust cleaning member **80** approaches the back side and the tip-end portion **91A** comes into contact with the back wall **84**, further movement of the holder **91** is stopped. In addition, the registration rollers **61A** and the movable frame **65**, which have been moved to the back side due to the contact friction force, also stop, and the registration rollers **61A** and the movable frame **65** are arranged at the initial position. In this state, when the front cover **111** is further closed and locked to the housing **11**, the mounting of the paper dust cleaning member **80** to the storage portion **81** is completed. By pushing the slide member **93** by the front cover **111** until the front cover **111** is locked to the housing **11**, the slide member **93** moves in a direction approaching the base end portion **91B** against the elastic force of the spring **97**. At this time, as shown in FIG. **11**, the hook portions **96A** are inserted into the slot holes **95A**; however, a state is maintained in which the hook portions **96A** are not engaged with edge portions of the slot holes **95A**.

(94) When the paper dust cleaning member **80** is mounted in the storage portion **81**, the elastic force with which the spring **97** biases the holder **91** is weaker than the contact friction force between the paper dust removing portion **90** and the registration roller **61A**. That is, in the present

embodiment, the spring **97** is applied such that the elastic force of the spring **97** is weaker than the contact friction force between the paper dust removing portion **90** and the registration roller **61A** in the mounted state.

(95) Note that the positioning mechanism of the present invention is configured by at least the front cover **111**, the spring **97**, the movable frame **65**, and the flat support portion **83B**.

(96) The operation of removing the paper dust cleaning member **80** from the storage portion **81** is performed in a state in which the power supply of the image forming apparatus **10** is turned OFF.

(97) When the front cover **111** (see FIG. **1**) of the image forming apparatus **10** is opened, the pressing by the front cover **111** is released. At this time, the paper dust cleaning member **80** moves from the contact position shown in FIG. **7** to the position shown in FIG. **14** due to the force generated when the slide member **93** slides forward due to the elastic force of the spring **97**. When the paper dust cleaning member **80** is pulled out a little, the paper dust cleaning member **80** is guided to the separated position shown in FIG. **13**. At this time, the paper dust removing portion **90** of the paper dust cleaning member **80** and the surface of the registration roller **61A** are separated by a predetermined gap. By pulling out the paper dust cleaning member **80** in this state, the paper dust cleaning member **80** can be pulled out from the storage portion **81** without bringing the paper dust removing portion **90** into sliding contact with the surface of the registration roller **61A**.

(98) [Control Portion **30**]

(99) The control portion **30** performs overall control of the image forming apparatus **10**, controls the operation of the sheet conveying unit **23** and the operation of the sheet correction unit **60**, and controls the conveying speed of each conveying roller pair. As shown in FIG. **4**, the control portion **30** includes a CPU **31**, a ROM **32**, a RAM **33**, a flash memory **34**, a motor driver **35**, and the like. The control portion **30** is an example of a movement control portion of the present invention. The control portion **30** is electrically connected to each motor **51**, **52**, **53** and each sensor **76**, **77**, **78** via a signal line or the like. Note that each of the motors **51**, **52**, and **53** is connected to a motor driver **35** of the control portion **30**, and is driven and controlled by receiving individual control signals from the motor driver **35**.

(100) In the present embodiment, the control portion **30** operates the sheet correction unit **60** to correct the skew of a sheet that is conveyed in a skewed manner.

(101) [Sheet Correction Process]

(102) Hereinafter, an example of the procedure of the sheet correction process executed by the control portion **30** will be described using the flowchart of FIG. **15**.

(103) Note that in the image forming apparatus **10**, it is assumed that the movable frame **65** of the sheet correction unit **60** is arranged at the home position before the sheet correction process is performed.

(104) When an instruction signal indicating the start of an image forming operation is input to the image forming apparatus **10**, the image forming apparatus **10** starts an image forming process. First, the control portion **30** starts conveying a sheet (**S11**). More specifically, in the control portion **30**, the motor driver **35** drives the conveying motors **51**, **53**, other motors (not shown), and the like, to rotate the drive roller **22A** of the feeding roller pair **22**, the pickup roller **29**, the drive roller **23A** of the sheet conveying unit **23**, the discharge roller pair, and the like. Thus, the sheet is taken out from the sheet storing portion **27** and fed to the conveying path **26**, and the sheet is conveyed in the conveying direction **D11** by the feeding roller pair **22** and the sheet conveying unit **23**.

(105) In the next step **S12**, the control portion **30** determines whether the leading edge of the sheet has been detected based on the detection signal output from the leading-edge detection sensor **77**. That is, the control portion **30**, by the leading edge detection sensor **77**, determines whether the leading edge of the sheet has passed the detection position.

(106) When it is determined in step **S12** that the leading edge of the sheet has been detected, the control portion **30** performs a registration operation to correct the skew of the sheet (**S13**). More specifically, the control portion **30** temporarily stops the registration roller **61A** before the sheet

enters the registration roller **61A**. Thus, the leading edge of the sheet abuts against a nip portion between the registration roller **61A** and the driven roller **61B**, and the skew of the sheet is corrected. Note that after the registration operation, the registration roller **61A** is driven again and the sheet is conveyed again.

(107) When the leading edge of the sheet reaches the edge detection sensor **78**, in the next step **S14**, the control portion **30**, based on the detection signal from the edge detection sensor **78**, determines whether the sheet has shifted in either direction (front side or rear side) in the width direction **D2**. Here, in a case where a shift (lateral shift) in the width direction **D2** has occurred, the process proceeds to step **S15**, and the shift direction and lateral shift amount are determined. On the other hand, in a case where no lateral shift has occurred, the process proceeds to step **S18**.

(108) In step **S16**, the control portion **30** calculates the direction of lateral shift of the sheet (shift direction) and the lateral shift amount. After that, in step **S17**, the control portion **30** slides the movable frame **65** of the sheet correction unit **60** in the direction of shift correction according to the lateral shift amount. More specifically, the control portion **30** drives and controls the correction motor **52** to move the movable frame **65** from the home position by the lateral shift amount in a direction opposite to the lateral shift direction of the sheet. Thus, the lateral shift of the sheet is corrected. After the sliding movement in step **S17** is performed, the process proceeds to step **S18**.

(109) In step **S18**, the control portion **30** determines whether the trailing edge of the sheet has passed through the sheet correction unit **60**. When it is determined that the trailing edge of the sheet has passed the sheet correction unit **60**, the control portion **30** returns the movable frame **65** to the home position (**S19**). When the process of forming an image on the sheet is completed, the control portion **30** stops driving each motor **51**, **52**, and **53**, and ends the series of processes.

(110) As described above, the image forming apparatus **10** of the present embodiment is provided with the paper dust cleaning member **80** having the paper dust removing portion **90**, and the storage portion **81** that slidably supports the holder **91** of the paper dust cleaning member **80**. Therefore, even in a case where the movable frame **65** is slid to correct the lateral shift and the registration rollers **61A** are also moved in the same direction, the holder **91** and the paper dust removing portion **90** are slid in the same direction by following the slide of the registration rollers **61A**. Thus, deterioration and damage to the paper dust removing portion **90** due to the axial sliding of the registration roller **61A** is reduced, and shortening of the life of the paper dust removing portion **90** is prevented. In addition, the paper dust removing portion **90** does not deform as the registration roller **61A** slides in the axial direction, and thus the removed paper dust inside the paper dust removing portion **90** is prevented from scattering.

(111) In addition, when power to the image forming apparatus **10** is OFF, the paper dust cleaning member **80** is inserted into the storage chamber **81A** of the storage portion **81** from the insertion opening **82**, and the paper dust cleaning member **80** is mounted to the mounting position at the back of the storage portion **81**, and then during the mounting process, the paper dust removing portion **90** and the surface of the registration roller **61A** come into contact with each other. At this time, the force in the insertion direction of the paper dust cleaning member **80**, due to contact friction, acts as a force to move the registration roller **61A** and the movable frame **65** to the rear side. Thus, the registration roller **61A** and the movable frame **65** move to the rear side due to the contact friction. As a result, when the paper dust cleaning member **80** is mounted, the movable frame **65** can be placed at the initial position.

(112) In addition, the first inclined guide **101** is provided in the storage portion **81** and the second inclined guide **102** is provided in the holder **91**, and thus by simply opening the front cover **111** and pulling out the paper dust cleaning member **80** a little, the paper dust removing portion **90** of the paper dust cleaning member **80** can be separated from the surface of the registration roller **61A**. In this case, contrary to when the paper dust cleaning member **80** is mounted, the base end portion **91B** of the holder **91** is separated from the surface of the registration roller **61** by the second inclined guide **102** before the tip-end portion **91A**, after which the tip-end portion **91A** is separated

from the surface of the registration roller **61** by the first inclined guide **101**, and then the paper dust cleaning member **80** is guided to the separated position. Thus, during the process of pulling out the paper dust cleaning member **80**, the paper dust removing portion **90** no longer comes into sliding contact with the surface of the registration roller **61A**, and thus the removed paper dust in the paper dust removing section **90** is prevented from adhering again to the surface of the registration roller **61A**.

(113) Note that in the embodiment described above, a configuration was described in which when the paper dust cleaning member **80** is mounted, the tip-end portion **91A** is pushed up toward the registration roller **61A** by the first inclined guide **101** before the base end portion **91B**, then following the tip-end portion **91A**, the base end portion **91B** is pushed up toward the registration roller **61A** side by the second inclined guide **102**, and when pulling out the paper dust cleaning member **80**, the base end portion **91B** is separated from the surface of the registration roller **61** by the second inclined guide **102** before the tip-end portion **91A**, and then following the base end portion **91B**, the tip-end portion **91A** separates from the surface of the registration roller **61** by the first inclined guide **101**; however, the present invention is not limited to such a configuration. For example, as a support mechanism for the paper dust cleaning member **80**, a configuration may be applied in which the inclination angles of the inclined surfaces of the first inclined guide **101** and the second inclined guide **101** are the same angle, and when pulling out the paper dust cleaning member **80**, in response to the paper dust cleaning member **80** being moved from the mounting position toward the insertion opening **82** side, the paper dust cleaning member **80** is moved in parallel to the separation position.

(114) In addition, in the above-described embodiment, a configuration was described in which the front cover **111**, as an example of a locking member, is closed in the housing **11**, thereby holding the slide member **93** in a pressed state; however, the present invention is not limited to such a configuration. Apart from the front cover **111**, a cover member that closes and locks the insertion opening **82** may be provided on the movable frame **65**. Moreover, instead of the cover member, for example, a locking mechanism may be provided that locks the slide member **93** in a state where the paper dust cleaning member **80** is mounted to the storage portion **81**.

(115) Furthermore, in the above-described embodiment, the paper dust cleaning member **80** for removing paper dust from the registration roller **61A** is illustrated; however, for example, the paper dust cleaning member **80** may be applied to removing paper dust from conveying rollers other than the registration roller **61A**, such as the drive roller **22A** and the drive roller **23A**. For example, in a case where play is provided in the axial direction of the rotating shafts of the drive rollers **22A**, **23A**, the drive rollers **22A**, **23A** may move in the axial direction due to vibrations generated during operation of the image forming apparatus **10**. In this case as well, the holder **91** and the paper dust removing portion **90** can be moved in the same direction following the axial movement of the drive rollers **22A**, **23A**.

(116) In addition, in the above-described embodiment, the image forming apparatus **10** is illustrated as an embodiment of the present invention; however, the present invention may also be viewed as a paper dust cleaning device configured by the storage portion **81**, the paper dust removing portion **90**, and the holder **91**.

(117) It is to be understood that the embodiments herein are illustrative and not restrictive, since the scope of the disclosure is defined by the appended claims rather than by the description preceding them, and all changes that fall within metes and bounds of the claims, or equivalence of such metes and bounds thereof are therefore intended to be embraced by the claims.

Claims

1. An image forming apparatus comprising a paper dust cleaning device configured to remove paper dust adhering to a surface of a conveying roller that conveys a sheet of paper in a

predetermined conveying direction; the paper dust cleaning device comprising: a paper dust removing member configured to extend in an axial direction of the conveying roller and come in contact with a surface of the conveying roller; a holder formed in a long shape in the axial direction and configured to hold the paper dust removing member; a holder support portion configured to hold the holder so that the paper dust removing member comes in contact with the surface of the conveying roller, and configured to support the holder so as to be movable in the axial direction; and an elastic member configured to apply an elastic force in the axial direction to the holder, the force being weaker than contact friction force between the surface of the conveying roller and the paper dust removing member.

2. The image forming apparatus according to claim 1, further comprising: a roller support frame configured to rotatably support the conveying roller; wherein the holder support portion is integrally formed with a roller support frame, and includes an insertion opening into which the holder is inserted, a support plate configured to extend along the axial direction from a lower end portion of the insertion opening and define a storage space on an upper side in which the holder is stored; and a contact portion provided on a back side of the storage space and against which a tip-end portion of the holder comes into contact.

3. The image forming apparatus according to claim 2, wherein the holder includes a slide member attached to a base end portion on one side in a longitudinal direction of the holder, the slide member forming a space between the slide member and the base end portion and being capable of sliding in the longitudinal direction; the elastic member is provided between the slide member and the base end portion so as to generate the elastic force; and the paper dust cleaning device further comprising a locking member configured to press the slide member toward the back of the storage space against the elastic force to hold the holder in the storage space in a state in which the holder is inserted into the storage space.

4. The image forming apparatus according to claim 3, wherein the locking member is a cover member that closes the insertion opening, and in a case where the insertion opening is closed by the cover member, a back surface of the cover member comes into contact with the slide member and presses the slide member toward a back side of the storage space.

5. The image forming apparatus according to claim 4, wherein the insertion opening is arranged on a predetermined side surface of the image forming apparatus; and the cover member, together with closing the insertion opening, also closes an opening on the side surface of the image forming apparatus.

6. The image forming apparatus according to claim 2, further comprising: a frame support portion configured to support the roller support frame so as to be movable in a width direction perpendicular to the conveying direction; a drive portion configured to supply a drive force in the width direction to the roller support frame; a positioning mechanism configured to move the roller support frame and position the roller support frame at a predetermined initial position; and a movement control portion configured to control the drive portion to move the roller support frame from the initial position to a predetermined home position when the image forming apparatus is energized; wherein the positioning mechanism is configured such that when power to the image forming apparatus is turned OFF, a force in an insertion direction of the holder generated when the holder is inserted into the storage space is transferred from the holder to the roller support frame via the conveying roller and moves the roller support frame to the initial position.

7. A paper dust cleaning device that removes paper dust adhered to a surface of a conveying roller that conveys a paper sheet in a predetermined conveying direction, comprising: a paper dust removing member configured to extend in an axial direction of the conveying roller and come in contact with a surface of the conveying roller; a holder configured to be long in the axial direction and hold the paper dust removing member; a holder support portion configured to hold the holder so that the paper dust removing member comes in contact with a surface of the conveying roller, and support the holder so as to be movable in the axial direction; and an elastic member configured

to apply an elastic force in the axial direction to the holder, the force being weaker than contact friction force between the surface of the conveying roller and the paper dust removing member.
